Intermediate Computer Vision Roadmap

1. CNN Basics

Learn how image AI works internally.

Topics:

- Convolution
- Filters/Kernels
- Feature maps
- Pooling
- Activation functions
- Fully Connected layers
- Forward propagation

Learn With

Computer Vision
Convolutional Neural Network

Practice Projects

- Cat vs Dog classifier
- Handwritten digit recognition
- Fruit classification

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2. Object Detection

Detect objects with bounding boxes.

Important Models

Model	Speed	Accuracy
YOLO	Very Fast	High

SSD	Fast	Medium
Faster R-CNN	Slow	Very High

Learn

OpenCV
YOLO
PyTorch
TensorFlow

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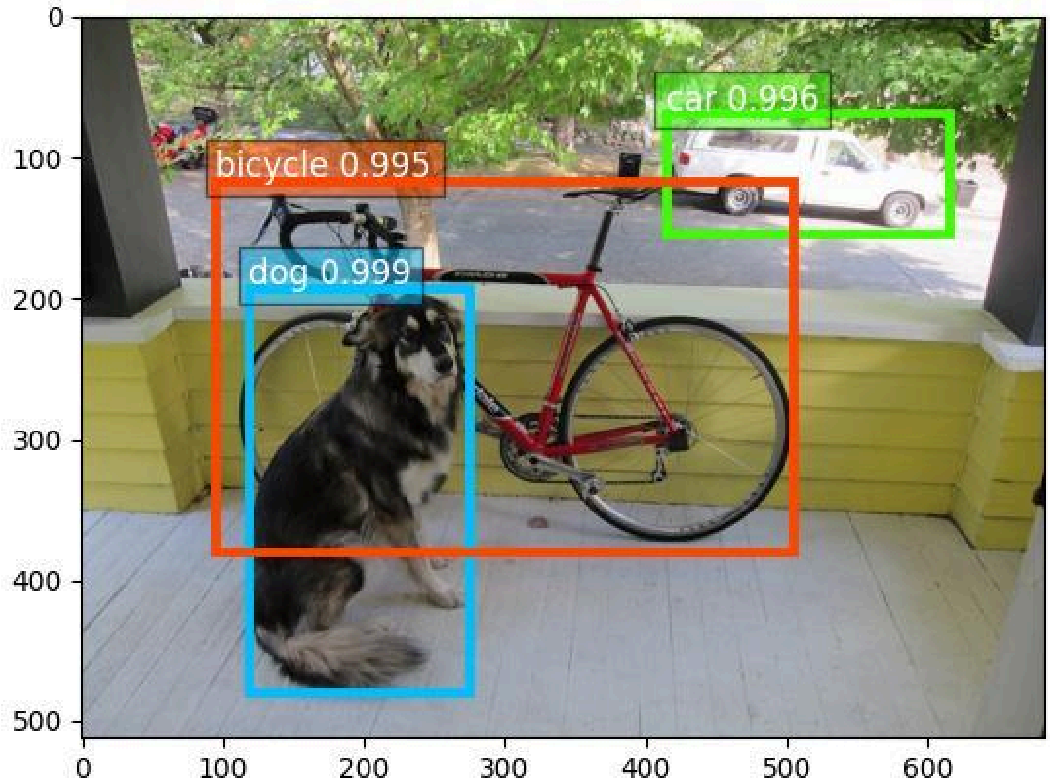
3. Object Tracking

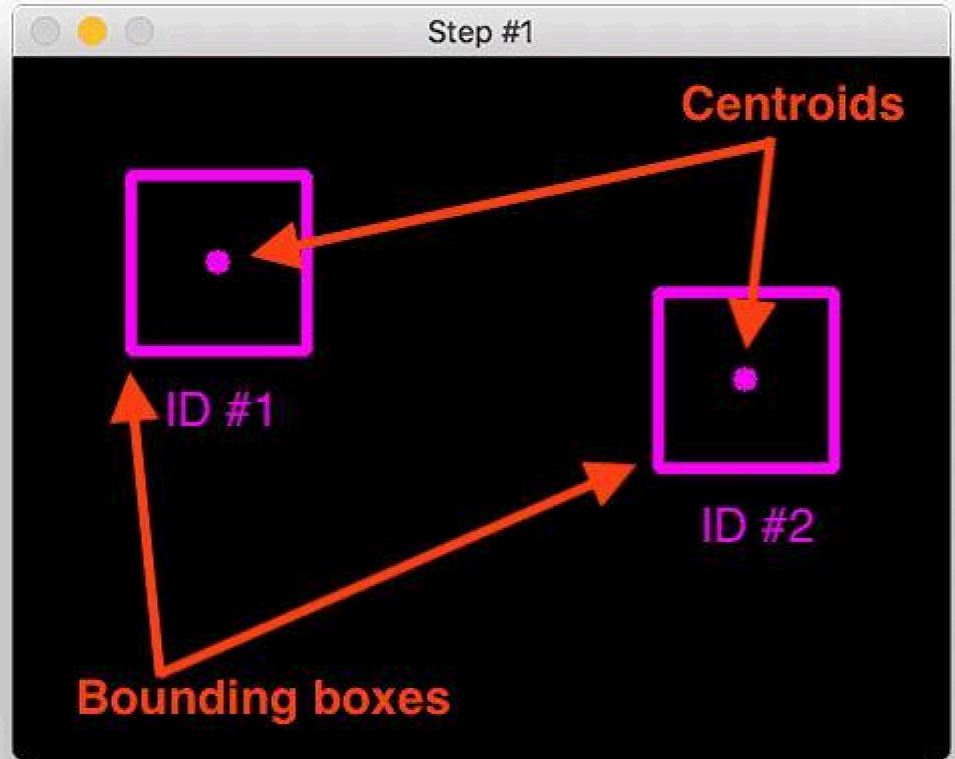
Track moving objects across video frames.

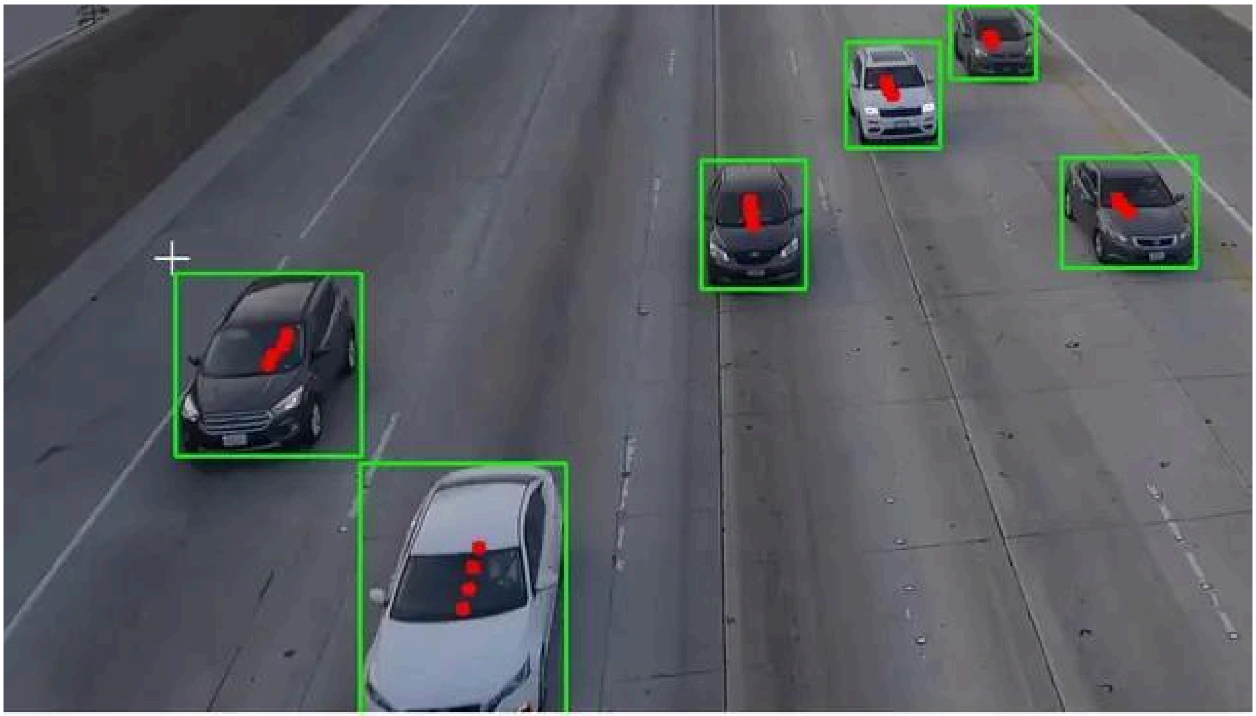
Algorithms

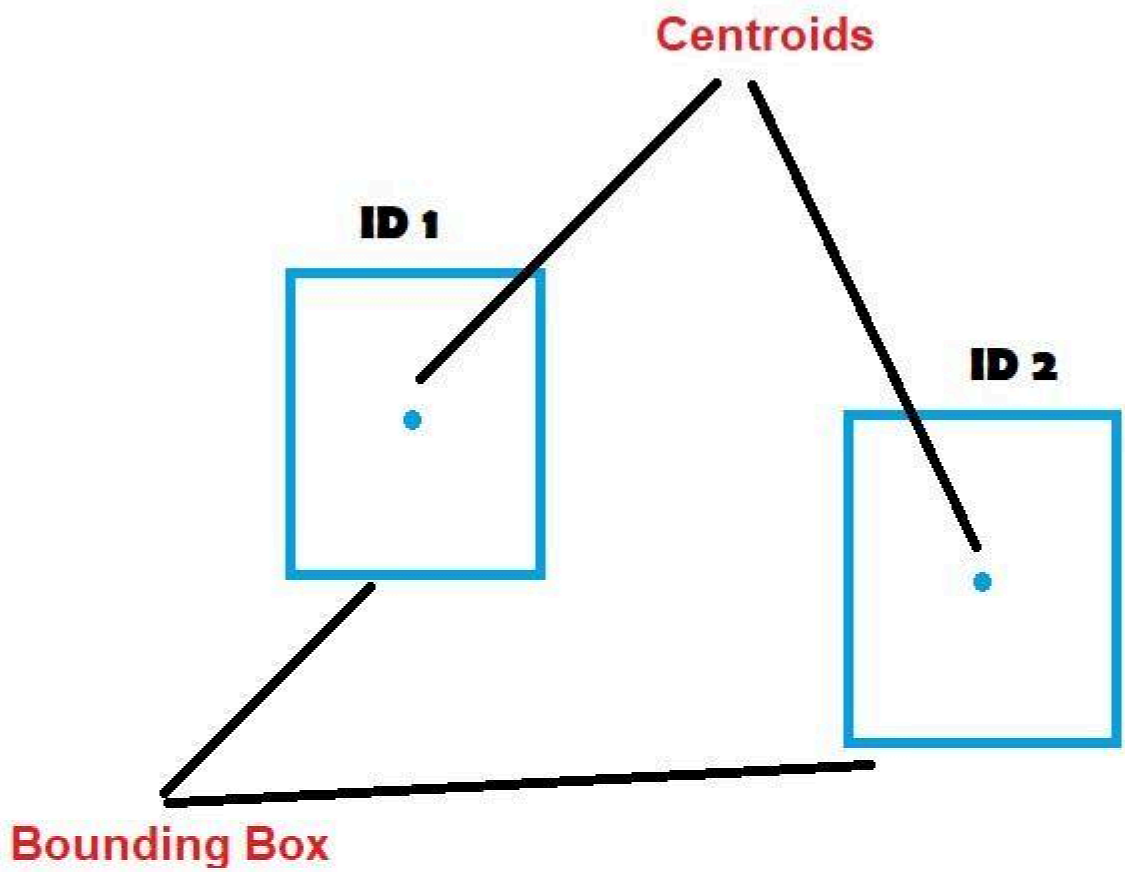
Tracking Method	Use
SORT	Simple tracking
DeepSORT	Better accuracy
Optical Flow	Motion tracking
ByteTrack	Modern fast tracker

Practice Projects

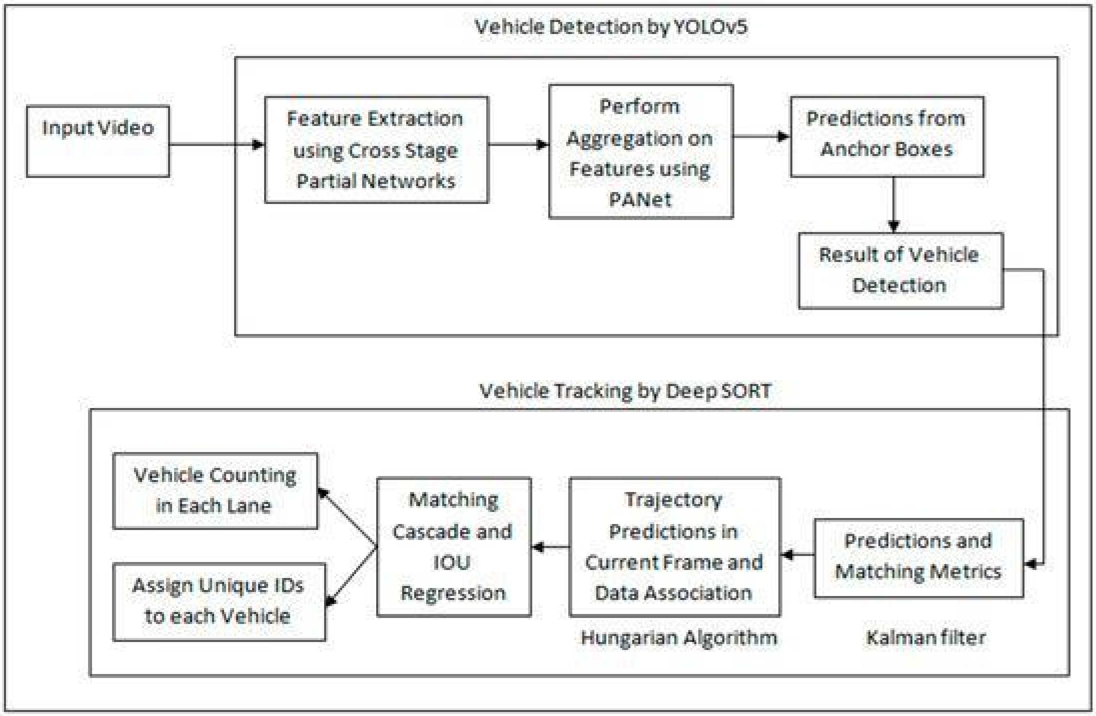

car 0.996
bicycle 0.995
dog 0.999
0
100
200
300
400
500
0
100
200
300
400
500
600

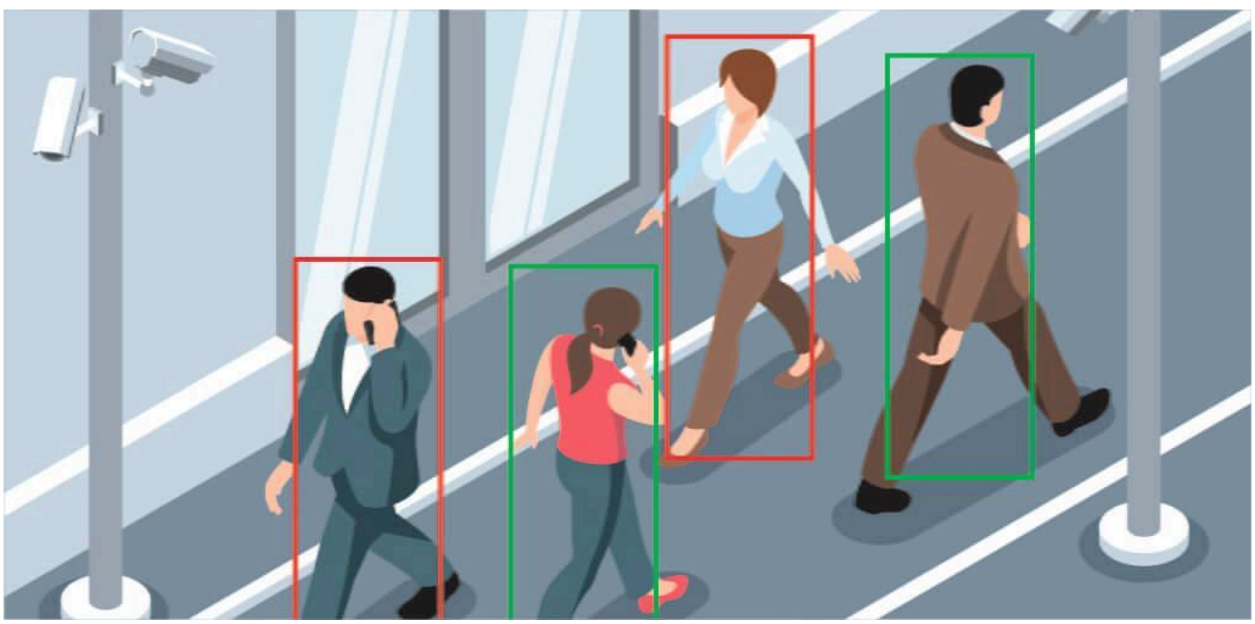

Step #1
Centroids
ID #1
ID #2
Bounding boxes

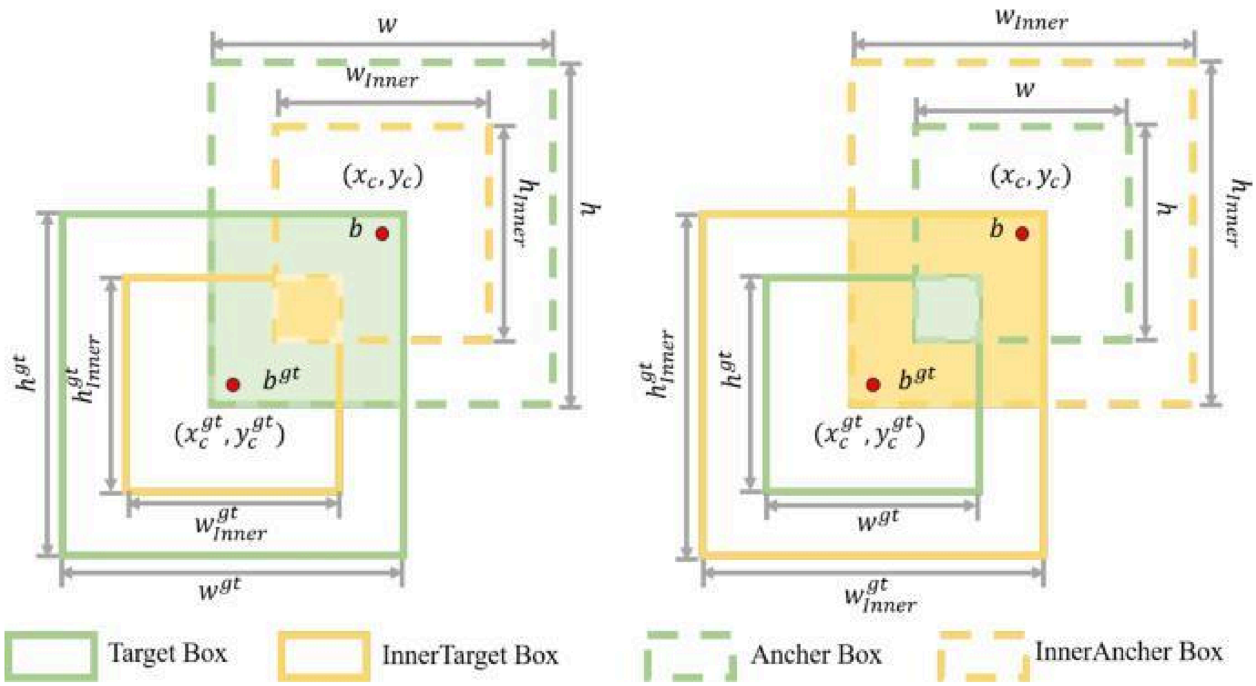




Centroids
ID 1
ID 2
Bounding Box


Vehicle Detection by YOLOv5
Input Video
Feature Extraction using Cross Stage Partial Networks
Perform Aggregation on Features using PANet
Predictions from Anchor Boxes
Result of Vehicle Detection
Vehicle Tracking by Deep SORT
Vehicle Counting in Each Lane
Assign Unique IDs to each Vehicle
Matching Cascade and IOU Regression
Trajectory Predictions in Current Frame and Data Association
Predictions and Matching Metrics
Hungarian Algorithm
Kalman filter





- vehicle counting
- people tracking
- face tracking
- drone tracking
- CCTV analytics

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4. Dataset Creation

Very important skill.

Tools

Tool	Purpose
LabelImg	Bounding box labeling
CVAT	Advanced annotation
Roboflow	Dataset management

Learn Dataset Formats

- YOLO format
- COCO format
- Pascal VOC

Example YOLO Dataset Structure

```
dataset/
 ├── images/
 │    ├── train/
 │    └── val/
 │
 ├── labels/
 │    ├── train/
 │    └── val/
 │
 └── data.yaml
```

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5. YOLO Training

Install

```
pip install ultralytics
```

Train Model

```
from ultralytics import YOLO

model = YOLO("yolov8n.pt")

model.train(
    data="data.yaml",
    epochs=50,
    imgsz=640
```

)

---

6. Real-Time Detection

Webcam Detection

```
from ultralytics import YOLO
import cv2

model = YOLO("best.pt")

cap = cv2.VideoCapture(0)

while True:

    ret, frame = cap.read()

    results = model(frame)

    annotated = results[0].plot()

    cv2.imshow("Detection", annotated)

    if cv2.waitKey(1) == 27:
        break
```

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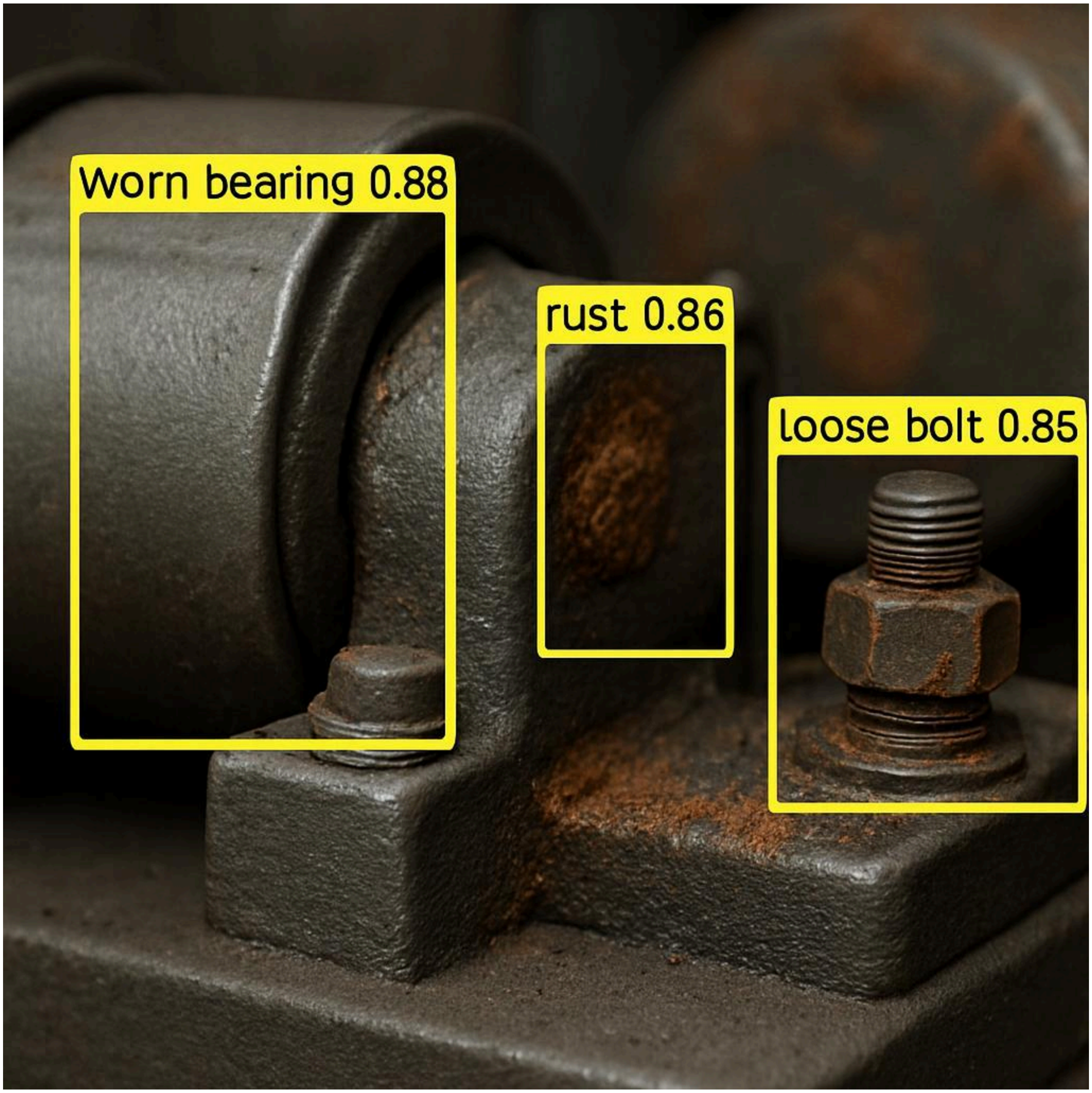
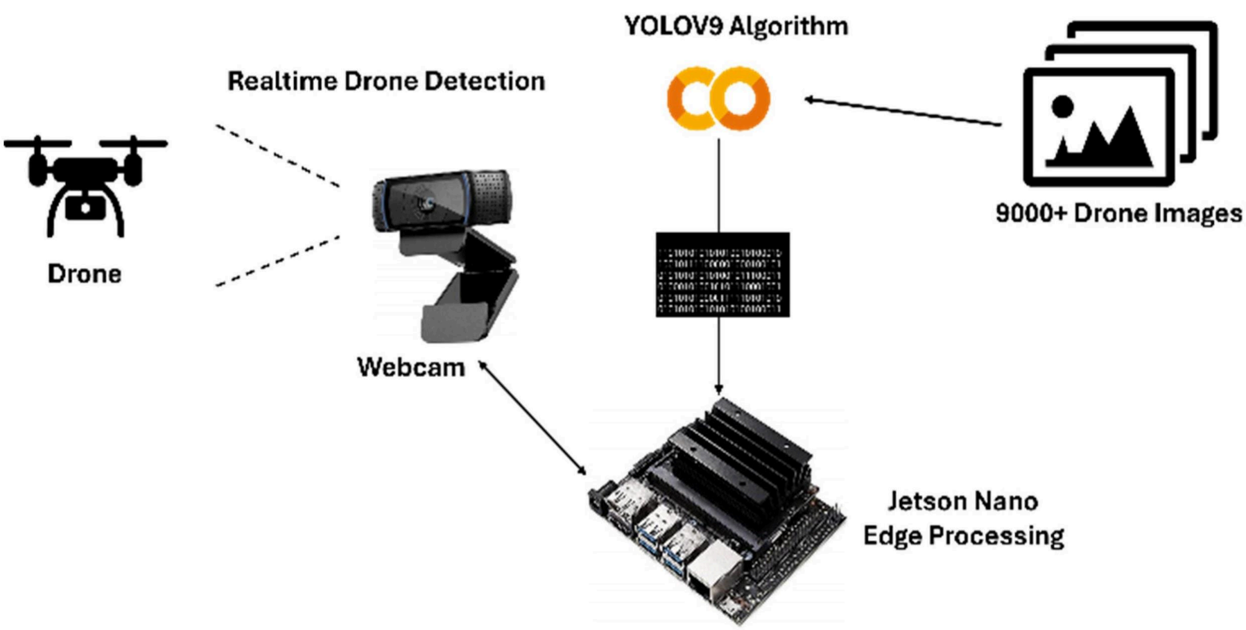
7. Intermediate Projects

(**c**)

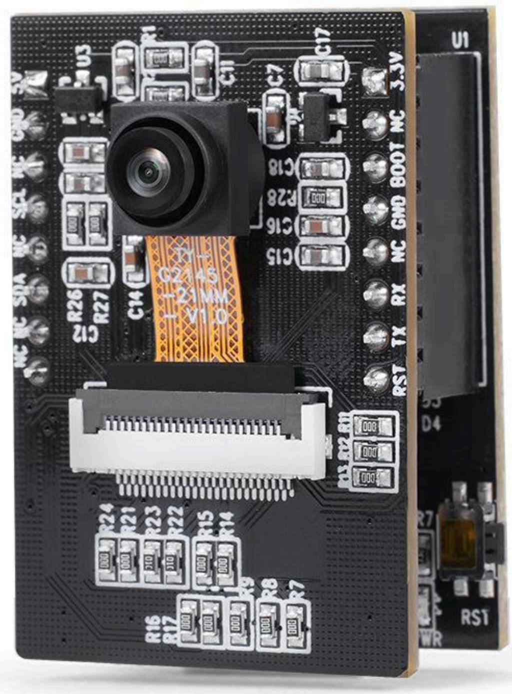
(**d**)

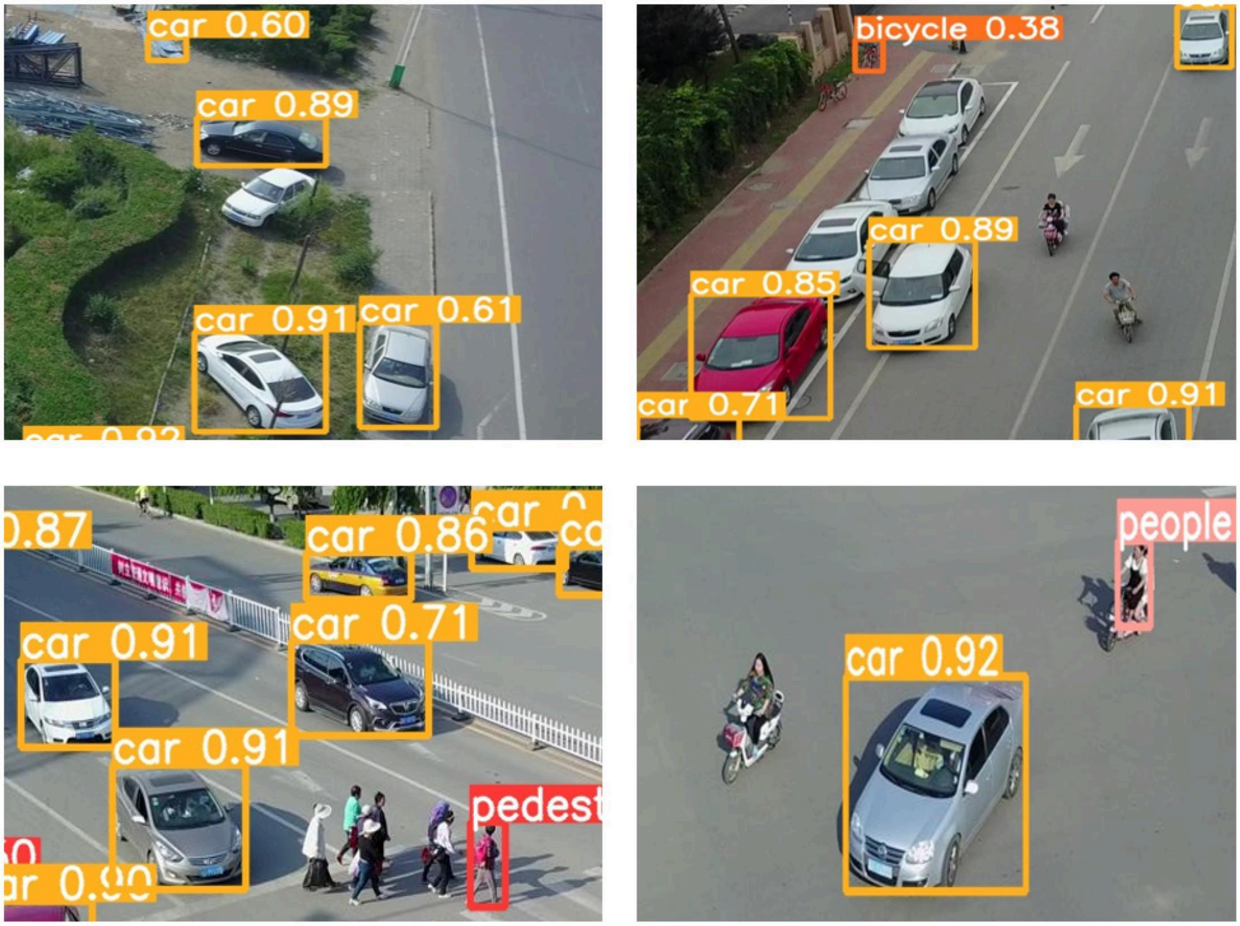

Worn bearing 0.88
rust 0.86
loose bolt 0.85

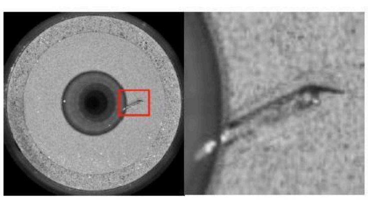

01010
DETECT


YOLOV9 Algorithm
Realtime Drone Detection
Drone
Webcam
9000+ Drone Images
Jetson Nano
Edge Processing

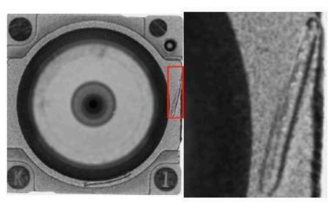
(a)


U1
3.3V
NC
BOOT
GND
NC
RX
TX
RST
RST

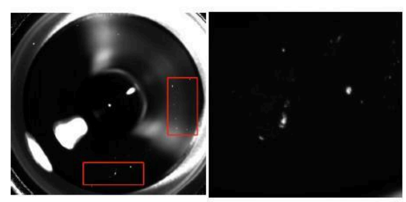




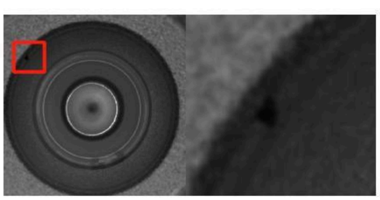
端面伤



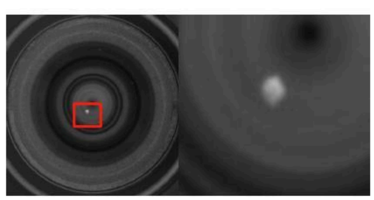
凸台伤



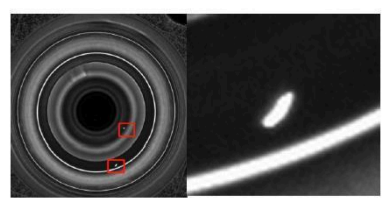
表面脏



黑点



脱模



白点

Good Projects

- AI CCTV
- Helmet detection
- Fire/smoke detection
- Number plate recognition
- Thermal camera AI
- Animal detection
- Industrial defect detection

Skills You Should Master

Skill	Importance
Python	Very High
OpenCV	Very High
YOLO	Very High
PyTorch	High
Linux	High
Dataset labeling	High
GPU training	Medium
